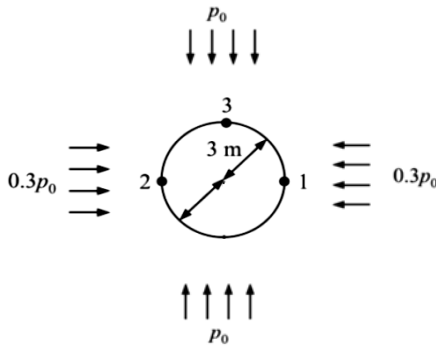


IV Semester B.Tech. (Mining Engineering)

Solutions for End semester Examination of
Rock Mechanics (MNC 205)Solution 1(a):10 marks

$$1.(a) \quad \sigma_v = \gamma z = 0.026 \times 450 = 11.7 \text{ MPa}$$

$$\sigma_h = 0.3 \times 11.7 = 3.51 \text{ MPa}$$

σ_{rr} is zero at three locations.

The tangential stress at points (1) & (2) is same because of symmetry.

$$\sigma_{\theta\theta} = \sigma_{1\theta\theta} = 31.59 \text{ MPa};$$

$$FOS = \frac{40}{31.59} = 1.27$$

at point (3), $\sigma_{\theta\theta} = -1.17 \text{ MPa}$

$$FOS = \frac{-2}{-1.17} = 1.71$$

For all locations, $FOS > 1$, hence stable.

Solution 1(b):10 marks

1.(b) At depth 800 m!

$$\sigma_h = p_s = 7 \text{ MPa}$$

$$\sigma_H = 3\sigma_h - p_B + \sigma_f = 3(7) - 14 + 10 = 17 \text{ MPa}$$

$$\sigma_v = 800 \times 0.026 = 20.8 \text{ MPa}$$

At depth 1200 m

$$\sigma_h = p_s = 19 \text{ MPa}$$

$$\sigma_H = 3(19) - 24.5 + 10 = 42.5 \text{ MPa}$$

$$\sigma_v = 1200 \times 0.026 = 31.2 \text{ MPa}$$

Solution 2(a): 5+5 = 10 marks

Applications of rock mass classification systems in mining and rock engineering

The applications of rock mass classification systems are:

- to identify the most significant parameters influencing the behaviour of a rock mass;
- to divide a particular rock mass formation into a number of rock mass classes of varying quality;
- to provide a basis for understanding the characteristics of each rock mass class;
- to derive quantitative data for engineering design;
- to recommend support guidelines for tunnels and mines;
- to provide a common basis for communication between engineers and geologists;
- to relate the experience on rock conditions at one site to the conditions encountered and experience gained at other.

Applications of Geomechanics Classification of rock masses (Rock Mass Rating (RMR)):

The classification may be used for estimating many useful parameters such as the unsupported span, the stand-up time or the bridge action period and the support pressure for an underground opening. It may also be used for selecting a method of excavation and the permanent support system. Further, cohesion, angle of internal friction, modulus of deformation of the rock mass and allowable bearing pressure for foundations may also be estimated for analysis of stability of rock slopes. Back-analysis of rock slopes in distress is more reliable approach for assessment of shear strength parameters. It also recommends cut slope angle along hill roads and rail lines.

Applications of Rock Mass Quality Index (Q System):

The Q-value may be used for classification of the rock mass around an underground opening, as well as for field mapping. This means that the Q-value depends on the underground opening and its geometry, and is therefore not an independent characterization of the rock mass. The Q-system is a classification system for rock masses with respect to stability of underground openings. Based on estimation of six rock mass parameters, a Q-value for a rock mass can be calculated. This value gives a description of the rock mass quality. The different Q-values are related to different types of permanent support by means of a schematic support chart. This means that by calculating the Q-value it is possible to find the type and quantity of support that has been applied previously in rock masses of the similar qualities. The Q-system can therefore be used as a guideline in rock support design decisions and for documentation of rock mass quality. The system can also be used for field mapping, core logging and investigations in a borehole.

Solution 2(b): 5 marks

$$2.(b) \quad RQD = \frac{\text{Sum of core pieces} \geq 10\text{cm}}{\text{total drill run}} \times 100 \%$$

$$= \frac{10.8 + 12.5 + 14.8 + 32.0 + 14.5 + 13.4}{150} \times 100 \%$$

$$= 65\% \text{ (approx.)}$$

$$\text{Core Recovery} = \frac{5.3 + 10.8 + 12.5 + 7.5 + 14.8 + 32.0 + 6.8 + 14.5 + 3.5 + 13.4}{150} \times 100$$
$$= 81\% \text{ (approx.)}$$

Solution 2(c): 5 marks

2.(c) plasticity Index, $PI = LL - PL$

$$PL = 90 - 50 = 40\%$$

$$\text{Liquidity Index} = \frac{W - PL}{LL - PL} = \frac{28 - 40}{50} = \frac{12}{50} = -0.24 (-ve)$$

semi-solid state of consistency & stiff

Since the liquidity index is negative, the soil is in the semi-solid state of consistency and is stiff.

Solution 3(a): 10 marks

3.(a) $\frac{\sigma_1}{\sigma_{ci}} = \frac{25}{50} = 0.5$

$$\frac{\sigma_3}{\sigma_{ci}} = \frac{5}{50} = 0.1$$

$$m_b = 10^e \left(\frac{42 - 100}{28} \right) = 1.26$$

$$S = e \left(\frac{42 - 100}{9} \right) = 0.00158$$

RHS of Hoek Brown failure criteria

$$= \frac{\sigma_3}{\sigma_{ci}} + \left(m_b \frac{\sigma_3}{\sigma_{ci}} + S \right)^{0.5}$$

$$= 0.1 + (1.26 \times 0.1 + 0.00158)^{0.5}$$

$$= 0.457$$

since $0.5 > 0.457$, the roof will fail.

Solution 3(b): 10 marks

Methods of reducing subsidence influence in mines:

Subsidence is the lowering of the ground surface following underground extraction of an orebody. Subsidence is produced, to a greater or a lesser degree, by almost all types of underground mining. Surface displacement may result from the redistribution of stresses associated with excavation or from mining-related activities such as de-watering.

Design measures to limit subsidence effects by the following methods:

- Partial extraction
- Goaf treatment
- Harmonic extraction
- Safety pillars
- Rapid Mining

Partial extraction involves leaving protective features such as pillars. The pillars are left between panels, has been successfully used to limit maximum subsidence. Depending on the lay out and the extraction ratio, reductions in maximum subsidence in order of 80% may be achieved. Partial extraction in which substantial rib pillars are left between panels, has been successfully used to limit maximum subsidence and to produce a net subsidence profile that is free from horizontal strain and tilt over most of its width.

Backfilling by hydraulic or pneumatic techniques, using a variety of materials including run-of-mine waste rock, milled tailings, sand, cemented backfilling material or other materials can reduce the subsidence up to 50% depending on the nature and timing of treatment. *Goaf treatment* by strip packing or hydraulic or pneumatic solid stowing can reduce the subsidence in a single panel by more than 50% depending on the nature and timing of the treatment. The largest reductions are obtained for solid stowing carried out immediately after mining.

Harmonic Mining involves the phased removal of the mineral from a critical area such that the ground surface is smoothly and horizontal strain is minimized. Harmonic extraction requires that the panel be advanced in at least two faces maintained at a carefully calculated distance apart. Harmonic extraction involves the phased removal of the mineral from a critical area such that the ground surface is lowered smoothly and horizontal strains are minimised. The technique may be used to protect structures that are especially important or susceptible to subsidence-induced damage. Harmonic extraction requires that the panel be advanced in at least two faces maintained at a carefully calculated distance apart. The orientation of the structure with respect to the direction of face advance determines whether protection against the transverse or the longitudinal surface wave is the more important.

Safety pillars are the solid pillars left untouched to protect surface structures located directly above. The size of the safety pillar can be determined by horizontal distance from a surface structure at which the advancing face must stop so that total strain at the site of the surface structure is less than the allowable values. The size of the safety pillar depends on the seam depth from the surface.

Rapid Mining: The maximum tensile strain (E_t) travelling with an advancing face is generally less than the final obtainable tensile strain, E_{max} and faster the mining, the less E_t is. Furthermore, the maximum compressive strain accompanied with the travelling face is always less than the corresponding maximum tensile strain, E_t . Therefore, there exists a minimum rate of mining characteristic of each coalfield that will induce maximum tensile traveling stain less than the allowable one.

Solution 4(a):10 marks

Monitoring underground excavation stability in hard rock mines using geotechnical instrumentation:

Monitoring is the surveillance of rock structures, either visually or with the help of instruments. Monitoring of rock mass performance in an excavation may be carried out for the following main reasons:

- to maintain safe operational procedure for personnel & equipment;
- to provide advance notice for instability or hazards;
- to get geotechnical information of rock or soil mass such as strength, deformability, anisotropy etc. for design/determination of supports and remedial technique;
- to analyse failure mechanism of rock or soil mass in which opening is constructed;
- to check change in hydrological behaviour of rock mass;

- identification of parameters causing instability of rock structures;
- identification of the change of stress in rock mass;
- measurement of the response of a rock mass to natural and man-made disturbance;
- to check the validity of the assumptions, conceptual models and values of soil or rock mass properties used in design calculations;
- to verify effectiveness of support and remedial measures.

Stability is a critical safety and production issue for mine structures. Major failure can occur seemingly without any visual warning, causing loss of lives, damage to equipment, and disruption to the mining process. Monitoring systems are employed to predict impending instabilities and failure.

Monitoring systems used in conjunction with underground mining operations can be ranging from simple to highly sophisticated and expensive. However, it should be remembered that valuable conclusions about rock mass response can often be reached from visual observations and from observations made using very simple monitoring devices. The parameters of a structure that need monitored are many but the most common ones are convergence/deformation, load, stress and ground water pressure.

Planning monitoring of rock mass

The following tasks are essential in systematic rock monitoring:

- Definition of benefits and purposes
- Planning of programme in accordance with rock conditions
- Selection of monitoring parameters
- Selection of instruments
- Writing of specifications for procurement and installation
- Procurement and installation
- Reading, maintenance and protection of instruments
- Data processing and presentation
- Interpretation
- Implementation

Selection of monitoring parameters:

Instrumentation is done after ascertaining what parameters to measure or detect.

Items that may be monitored in an underground mining operation include

- change in load on support
- change in static stress in rock mass
- measurement of displacement in excavation
- wall- wall and roof-floor convergence
- subsurface rock displacement
- change in hydrological properties in rock mass
- elevation of phreatic surface (gw)
- water pressure
- water flow
- ground vibration & dynamic stress in rock mass
- seismic/micro-seismic/acoustic behaviour of rock mass

Monitoring of rock mass

The parameters of a structure that need monitored are many but the most common ones are convergence/deformation, load, stress and ground water pressure.

Convergence monitoring on boundary of excavation

Convergence, or the relative displacement of two points on the boundary of an excavation, is probably the most frequently made underground measurement. The measurement is made variously with a telescopic rod, invar bar or tape under constant tension, placed between two measuring points firmly fixed to the rock surface. A dial gauge, micrometer, or an electrical device such as an LVDT, potentiometer or vibrating wire gauge is used to obtain the measurement of relative displacement. When interpreting convergence measurement results, it is important to consider the time rates of convergence and any changes in those rates, as well as the total convergence values.

A list of some geotechnical instruments for typical measurement is given below:

Measuring parameters	Instruments
1. Roof – Floor & Wall-wall Displacement/Deformation	• Convergence • Indicator/Recorder • Tape/Tube Extensometer • Rod/Wire • Tell – Tale • Surveying instrument (Profilometer, Distomat, Levelling Eqp.)
2. Displacement/Deformation within Rock Mass	• Rod/Wire borehole Extensometer • Vibrating wire borehole Extensometer • Potentiometer borehole Extensometer • Magnetic borehole Extensometer • Sonic borehole Extensometer • Inclinator • Shear strip • TDR-Time-Domain Reflectometry
3. Borehole Visual Inspecting Device	• Fiberoptic Bore Scope • Video Bore Scope • Bore Hole Television Camera
4. Load on Support	*Load Cell (Strain gauge/Hydraulic/Vibrating Wire/Potentiometer based) * Central Hole Load Cell (for Rock bolt) * Instrumented Rock Bolt
4. Stress Change in Rock mass	• Borehole Pressure Cell • Vibrating Wire Stressmeter (Stress Cell)
5. Ground Water level	• Open pipe Piezometer
6. Ground Water Pressure	• Piezometer (Pneumatic/Hydraulic/VW)

Solution 4(b): 10 marks

4.(b)

$$\begin{aligned}\text{Volume of the block, } V &= \left(\frac{H^2 b}{2}\right) [\cot \alpha - \cot \beta] \\ &= \frac{25^2}{2} (\cot 35^\circ - \cot 45^\circ) \\ &= 133.8 \text{ m}^3\end{aligned}$$

$$\begin{aligned}\text{Weight of the block, } W &= \rho V = 24.5 \times 133.8 \\ &= 3278 \text{ kN}\end{aligned}$$

$$\begin{aligned}\text{Driving force} &= W \sin \alpha = 3278 \times \sin 35^\circ \\ &= 1878.46 \text{ kN} \\ &= 1.878 \text{ MN}\end{aligned}$$

$$\begin{aligned}\text{Total cohesion, } C &= cA \\ &= c \times \frac{H}{\sin \alpha} = 0.02 \times \frac{25}{\sin 35^\circ} = 0.8717 \text{ MN}\end{aligned}$$

$$\begin{aligned}\text{Resisting force} &= \sigma_n \tan \phi + C \\ \left[\sigma_n &= 3278 \times \cos 35^\circ = 2685.18 \text{ kN} \right. \\ &\quad \left. = 2.685 \text{ MN} \right]\end{aligned}$$

$$\begin{aligned}\Rightarrow & 2.68518 \times \tan 30^\circ + 0.8717 \\ \Rightarrow & 2.42 \text{ MN}\end{aligned}$$

$$\text{FOS} = \frac{2.42}{1.88} = 1.3 \text{ (approx.)}$$

Solution 5 (a): 5 marks

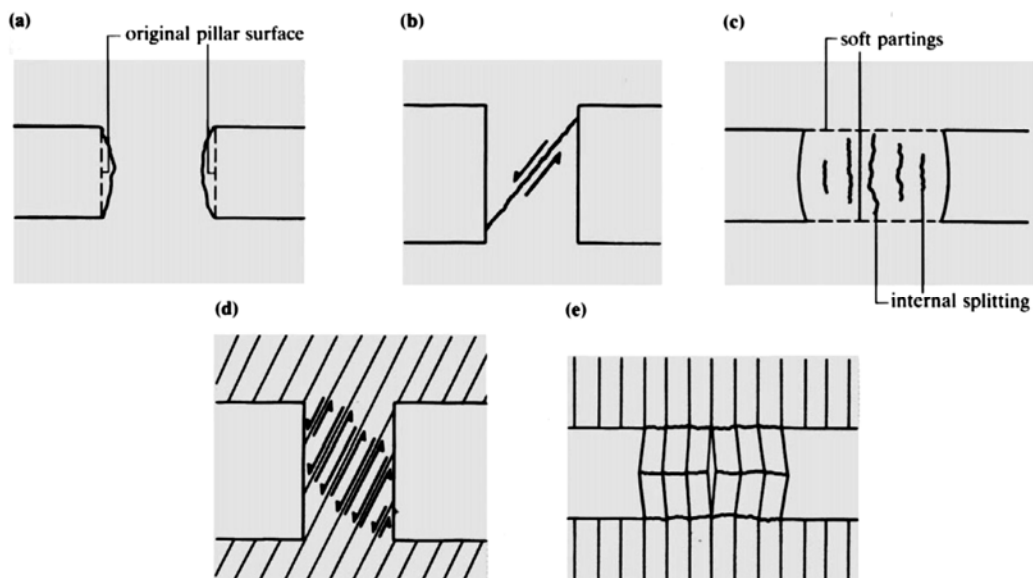


Fig: Principal modes of deformation behaviour of mine pillars.

- (a) Fretting or necking of pillars (b) Inclined shear failure (c) Internal splitting failure
(d) Structurally controlled failure (e) Structurally controlled buckling failure

In relatively massive rock, the most obvious sign of pillar stressing involves spalling from the pillar surfaces, as illustrated in Figure a. Fretting or necking of the pillar occurs.

The effect of pillar relative dimensions on failure mode is illustrated in Figure b. For regularly jointed orebody rock, a high pillar height/width ratio may favour the formation of inclined shear fractures transecting the pillar. There are clearly kinematic factors promoting the development of penetrative, localised shear zones of this type.

Yield of the soft layers generates transverse tractions over the pillar end surfaces and promotes internal axial splitting of the pillar. This may be observed physically as lateral bulging or barrelling of the pillar surfaces. Geomechanical conditions favouring this mode of response may occur in stratiform orebodies, where soft bedding plane partings define the foot wall and hanging wall for the orebody. The failure condition is illustrated in Figure c.

Other specific modes of pillar response may be related directly to the structural geology of the pillar. For example, a pillar with a set of natural transgressive fractures, as illustrated in Figure d, can be expected to yield if the angle of inclination of the fractures to the pillar principal plane (that perpendicular to the pillar axis) exceeds their effective angle of friction.

A pillar with a well-developed foliation or schistosity parallel to the principal axis of loading will fail in a buckling mode, as illustrated in Figure e. This mechanism resembles the formation of kink bands.

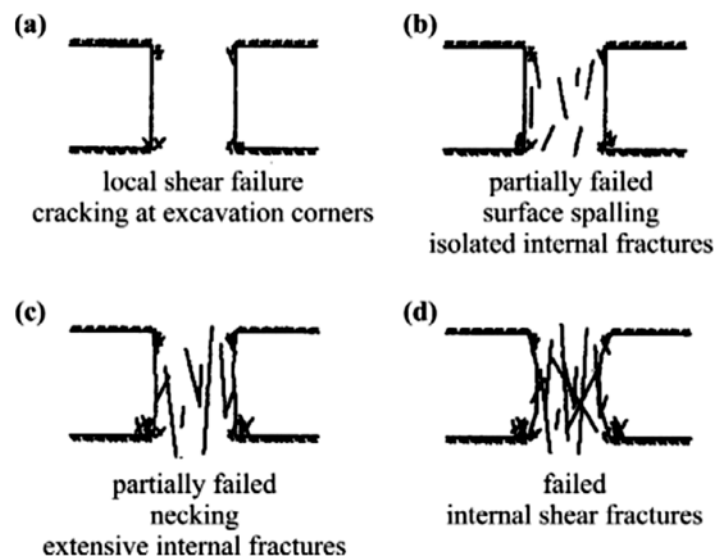


Fig: Schematic illustration of the evolution of fracture and failure in a pillar in massive rock

Solution 5 (b): 10 marks

5.(b)

Coal seam thickness = 3.1 m

depth = 80 m

extracted span, $w_0 = 5$ m

pillar size, $w_p = 7$ m square pillars.

extracted height, $h = 2.2$ m

$$\sigma_v = 0.026 \times 80 = 2.08 \text{ MPa}$$

Case(i), $h = 2.2$ m

$$\begin{aligned} \text{strength of pillar, } S &= 7.5 (2.2)^{-0.66} \times (7)^{0.46} \\ &= 10.91 \text{ MPa} \end{aligned}$$

$$\sigma_p = \sigma_v \left[\frac{w_0 + w_p}{w_p} \right]^2 = 2.08 \left[\frac{5+7}{7} \right]^2 = 6.11 \text{ MPa}$$

$$\text{FOS} = \frac{S}{\sigma_p} = \frac{10.91}{6.11} = 1.785$$

Case(ii) $h = 2.2 + 0.6 = 2.8$ m

$$S = 7.5 (2.8)^{-0.66} \times (7)^{0.46} = 9.3 \text{ MPa}$$

$$\text{FOS} = \frac{S}{\sigma_p} = \frac{9.3}{6.11} = 1.522$$

extra 0.6 m additional coal from the roof is feasible
since the FOS of pillars is 1.522, which is safe & stable.

Solution 5 (c): 5 marks

5. (c) For hydrostatic stress field, $K = \frac{\sigma_h}{\sigma_v} = 1$

$$\text{radial stress, } \sigma_{rr} = P \left[\left(1 - \frac{a^2}{r^2} \right) \right] \quad \because [P = 5 \text{ MPa}]$$

$$\text{For circular opening, C, } \sigma_{rr} = 5 \left[\left(1 - \frac{2.5^2}{5^2} \right) \right] = 3.75 \text{ MPa}$$

$$\text{For " " D, } \sigma_{rr} = 5 \left[\left(1 - \frac{2.5^2}{5^2} \right) \right] = 3.75 \text{ MPa}$$

At point A, there is ~~an~~ influence zone of influence due to two circular openings, C & D,

$$\sigma_{rr} = 3.75 + 3.75 = \underline{7.5 \text{ MPa}}$$

Solution 6 (a): 4+4 = 8 marks

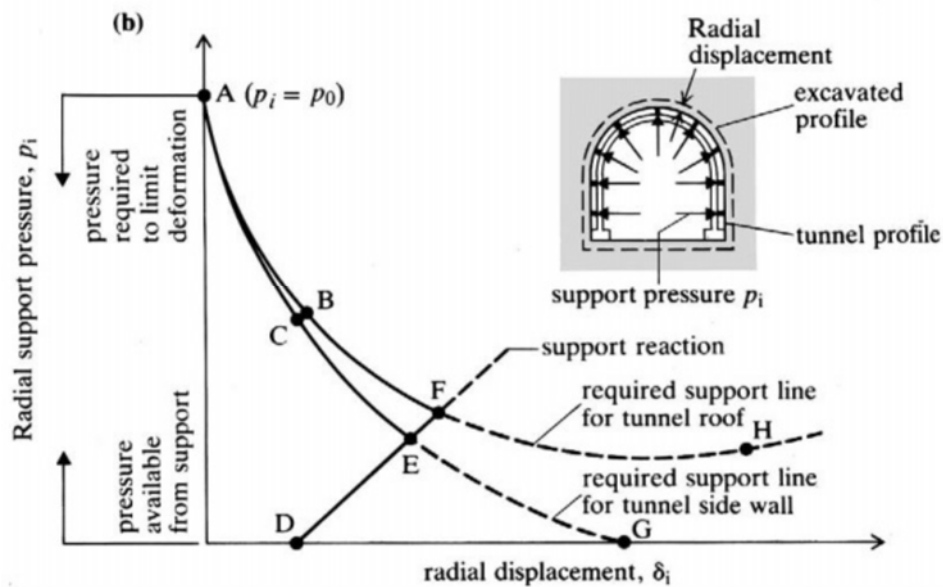


Fig: The radial support pressure–displacement curves for the rock mass and the support system

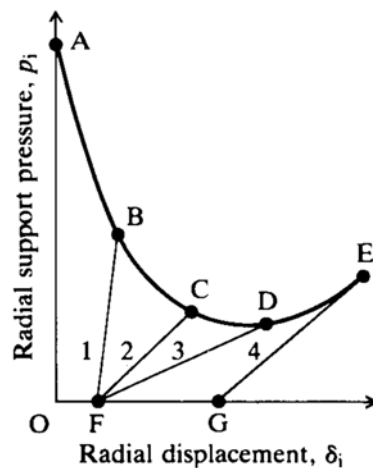


Fig: Illustration of the influence of support stiffness and of the timing of its installation on support performance.

Figure shows a rock–support interaction diagram. The ground characteristic or required support line is given by ABCDE. The earliest practicable time at which support can be installed is after radial displacement of an amount OF has occurred.

Support 1 is installed at F and reaches equilibrium with the rock mass at point B. This support is too stiff for the purpose and attracts an excessive share of the redistributed load. As a consequence, the support elements may fail causing catastrophic failure of the rock surrounding the excavation.

Support 2, having a lower stiffness, is installed at F and reaches equilibrium with the rock mass at C. Provided the corresponding displacement of the periphery of the excavation is acceptable operationally, this system provides a good solution.

Support 3, having a much lower stiffness than support 2, is also installed at F but reaches equilibrium with the rock mass at D where the rock mass has started to loosen.

Support 4, of the same type and stiffness as support 2, is not installed until a radial displacement of the rock mass of OG has occurred. In this case, the support is installed too late, excessive convergence of the excavation will occur, and the support elements will probably become overstressed before equilibrium is reached.

Solution 6 (b): 3+3 = 6 marks

6. (b)

$$\sigma_n = \frac{\sigma_x + \sigma_y}{2} + \frac{\sigma_x - \sigma_y}{2} \cos 2\theta + \tau_{xy} \sin 2\theta$$

$$\sigma_n = 25 + 10 \times \cos 80^\circ + 9 \times \sin 80^\circ = 35.6 \text{ MN/m}^2$$

$$\tau = \frac{\sigma_x - \sigma_y}{2} \sin 2\theta - \tau_{xy} \cos 2\theta$$

$$\tau = 10 \times \sin 80^\circ - 9 \times \cos 80^\circ = 8.29 \text{ MN/m}^2$$

$$\sigma_r = \sqrt{\sigma_n^2 + \tau^2} = \sqrt{35.6^2 + 8.29^2} = 35.55 \text{ MN/m}^2$$

$$\phi = \tan^{-1} \tau / \sigma_n = \tan^{-1} \frac{8.29}{35.6} = 13^\circ$$

Solution 6 (c): 6 marks

Comparison of FDM and FEM for rock mechanics applications

In practice finite element method is usually indistinguishable from the finite difference method. In these methods, relate the conditions at few points within the medium (nodal points) to the state within a finite closed region formed by these points (element). The physical problem is being modelled numerically by discretizing (i.e. dividing into zones or elements) the problem region.

The FDM is a direct approximation of the governing partial differential equations (PDEs) by replacing partial derivatives with differences at regular or irregular grids imposed over problem domains, transferring the original PDEs into a system of algebraic equations in terms of unknowns at grid points. The solution of the system equation is obtained after imposing necessary initial and boundary conditions. This method is the oldest member in the family of numerical methods, one that is widely applied.

In FEM, trial functions, usually polynomial, are used to approximate the behaviour of PDEs at the element level and generate the local algebraic equations representing the behaviour of the elements. The local elemental equations are then assembled, according to the topologic relations between the nodes and elements, into a global system of algebraic equations whose solution then produces the required information in the solution region, after imposing the properly defined initial and boundary conditions.

Finite Differential techniques (FDM) are far more powerful and versatile. Complicated problems involving complex geo-mining and stress environments, as often found in actual mining complexes, can be studied and solved in much more elaborate and realistic fashions in these methods. FDM results into

conditionally stable solution. That is, the convergence of the solution at different stages of iteration to a true solution depends on the size of elements and size of the load steps. It has also got the advantage of time-stepping which allows a better understanding of the trend and mode of a failure.

FEM is perhaps the most versatile of all methods and capable of yielding the most realistic results even in complex geo-mining conditions. Complexity in problem formulation and requirements of long computer time and large memory space seem to be its major shortcomings.

However, each method has its own advantages and disadvantages. The following table shows different features of each modelling method.

Finite Difference Method	Finite Element Method
method is an approximation to the differential equation	method is an approximation to the solution of the differential equation
solves a problem on a set of points that form a grid	solves a problem on the interiors of the grid cells (elements) and for the grid points
Easier to implement, but approximation between grid points can be problematic	Can more easily handle complex geometries
Uses regular grid	Can use irregular grid
Incremental time steps need to be smaller than critical value for stability	Time steps can be large with unconditionally stable schemes
Small amount of calculations in each time step	Large amount of calculations in each time step
No iterations necessary to follow nonlinear constitute law	To follow a nonlinear constitute law an iterative procedure is required
Following of nonlinear laws in a valid physical way is guaranteed	The following of physically correct path should always be assured
Low memory requirement hence no matrices are formed	Large memory requirement for storage of stiffness matrices
